

Agenda - 15-02-2025

- RNN
- LSTM

Embedding - Recap

① ② ③ ④ ⑤
king, queen, person, man, woman

ONE -

1x5

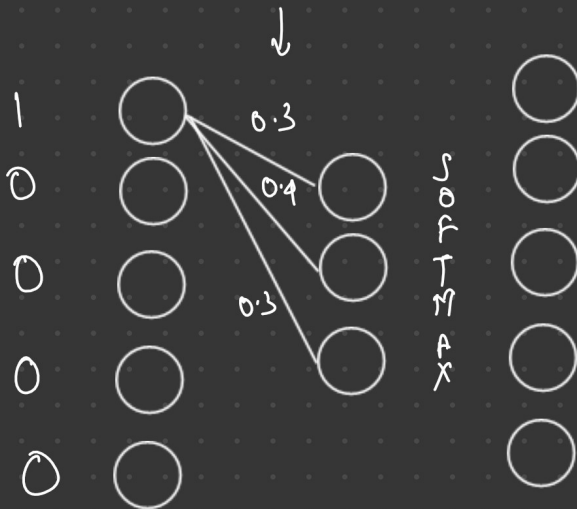
5x5

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$\rightarrow N=3$

[5x3]

↓ ↓ ↓
① - 0.3, 0.4, 0.5
② - 0.3, 0.4, 0.5
③ - 0.4, 0.5, 0.6
④ - 0.7, 0.8, 0.96
⑤ - 0.12, 0.16, 0.46



Neural Network :-

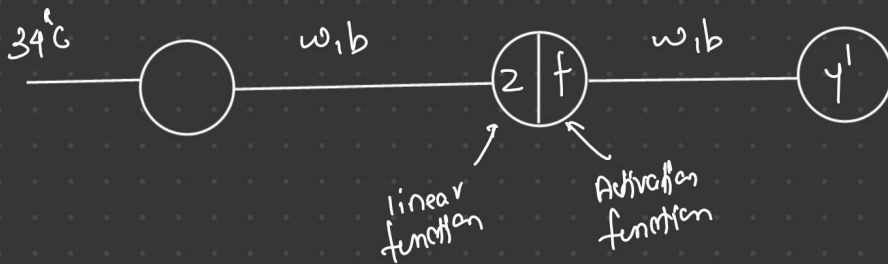
problem :- we have to predict tomorrow's temperature

Time-Series :-

Yesterday — Tomorrow
 $y-1, y \rightarrow$ Tomorrow
 $y-2, y-1, y \rightarrow$ Tomorrow
Historical data

Trends

$T \rightarrow$ Date
 — Day
 — 2023/2024/
 2025
 — month



$$w_1 = 1$$
$$b_1 = 0$$

$$w_2 = 30$$
$$b_2 = 0$$

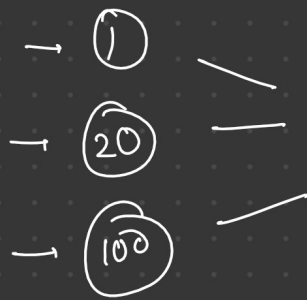
$$y' = \sigma^1 \cdot w^2 + b_1$$
$$= 0.99 \times 30 + 0$$
$$= \underline{\underline{29.7}}$$

$$z = w \times t + b$$
$$= 1 \times 34 + b$$
$$= 34$$
$$\sigma^1 = \sigma(34)$$
$$= 0.99$$

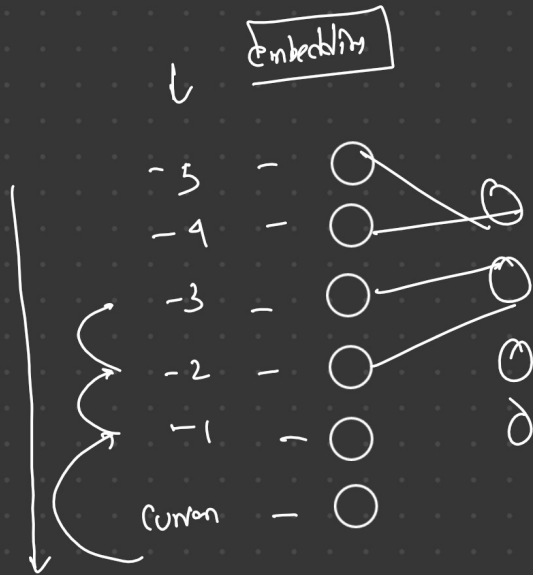
— Does not take Historical data into consideration

Input $\rightarrow 30$
hidden -1 $\rightarrow 100$
 -2 $\rightarrow 200$
 -3 $\rightarrow$ output layer

predict
Day -1 $\rightarrow$ ① T_{t-1}
 2 $\rightarrow$ ② T_t
 3 $\rightarrow$ ③
 365 $\rightarrow$ ③65



NN → variable length



Hi How are you

How are you Hi

Hi are

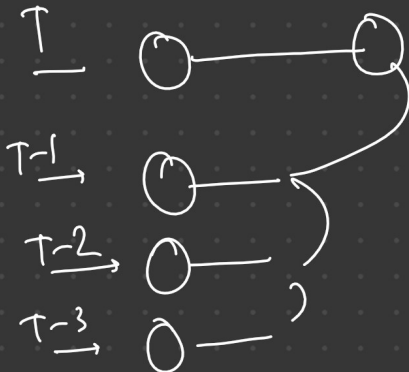
sequence

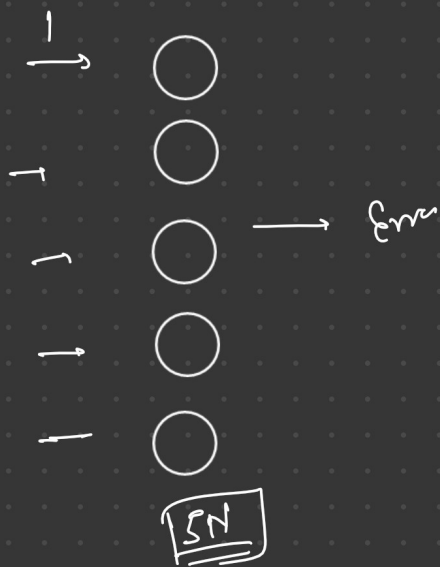
H_i

H_{i+1}

→ variable input

→ memory



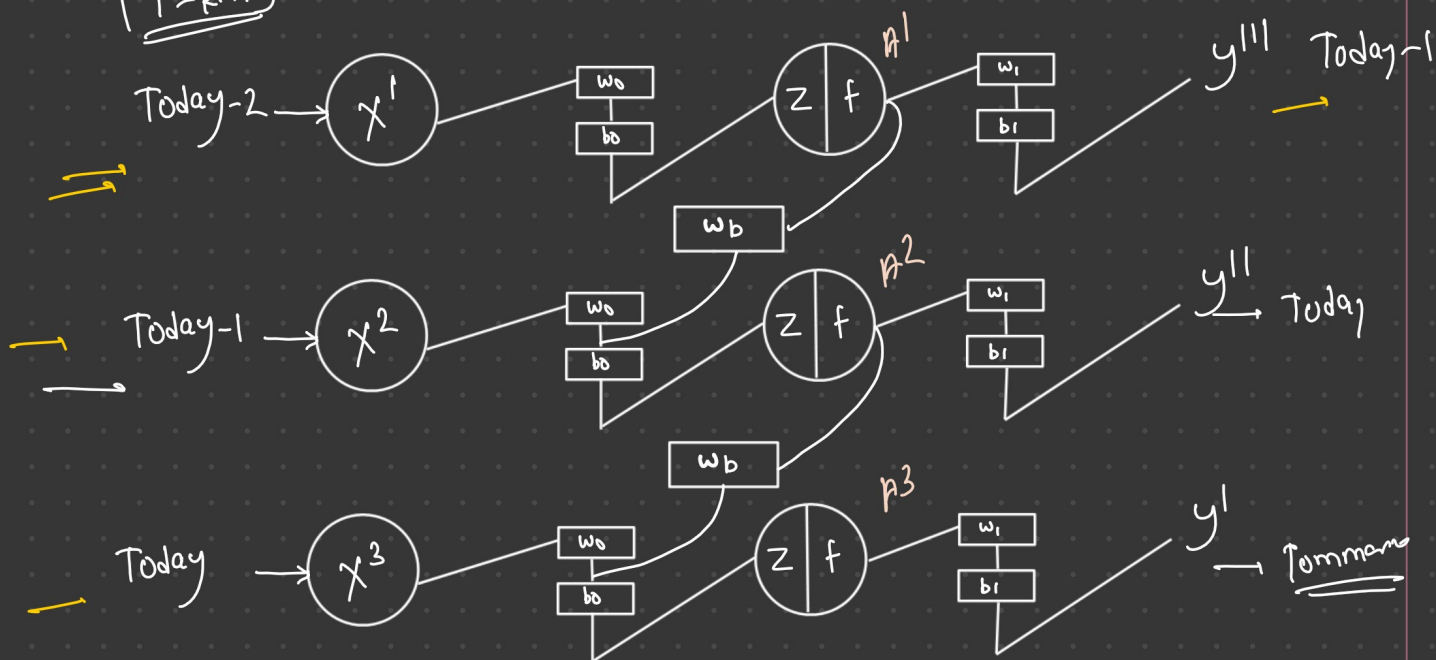


$\boxed{hi}$ — chatGPT — Yes
 $\boxed{Hi How are you}$ — Yes



1 — Token size

1-RNN



input — 100 days — RNN

11

$$\text{relu}(w x^1 + b) \rightarrow A^1$$

xx 0.001 x 0.001 x 0.001
small number

$$\text{relu}(w x^2 + \boxed{w_b^1 \cdot A^1} + b) \rightarrow A^2$$

$$\text{relu}\left(\left(w x^3 + \boxed{w_b^2 \cdot A^2}\right) + b\right) \rightarrow A^3 \rightarrow \text{Tomorrow}$$

loop 1.2 x 1.2 x 1.3

$$y^1 = w \cdot A^3 + b$$

4

xx 2 x 2 x 2 x 2
large number

$$w_b = 2$$

~~xx~~ 2 ↘

2 ↘

↘

↘

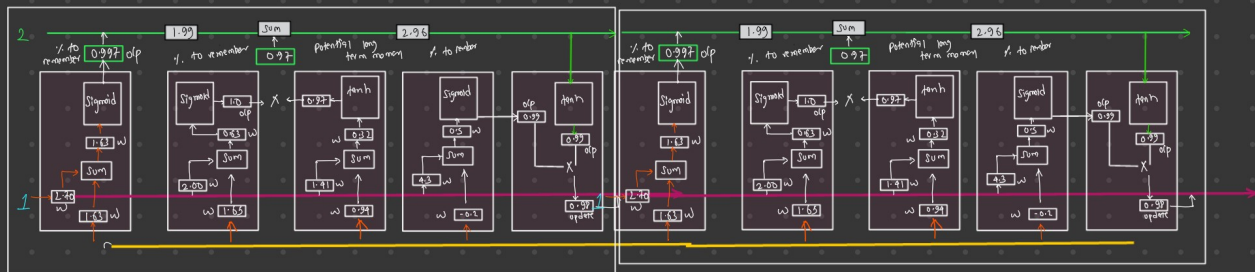
→ A^3 →

→ short term memory

Block 1 $\rightarrow$ % to remember long term memory (forget gate)

Block 2 $\rightarrow$ % to remember & update long term memory (Input Gate)

Block 3 $\rightarrow$ % to update short term memory (output gate)



$\uparrow$ 30

It

32

It - 2